higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MATERIALS
NQF LEVEL 3

(12010273)

14 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 6 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Use only BLUE or BLACK ink.
6. Write neatly and legibly.

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QUESTION 1

1.1 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1.1 Sapwood is the centre of the tree trunk.

1.1.2 The term *timber* is given to a tree after it has been sawn.

1.1.3 A tree that bears cones is an angiosperm.

1.1.4 Hardwood can be used to make furniture and paper and can even be used in construction.

1.1.5 Felling is a process of cutting down trees.

1.1.6 Structural timber is isotropic material.

1.1.7 Distortion is the breaking of timber into two pieces.

1.1.8 A knot is the area where the tree grew a branch that was later cut off.

1.1.9 Early wood is represented by the dark area on the growth ring.

1.1.10 Seasoning is when we remove excess moisture from wood.

(10 × 1) (10)

1.2 Show by means of a sketch how the following information appears on timber:

1.2.1 Manufacturer's name or trademark

1.2.2 The verification mark

1.2.3 Method of grading

1.2.4 Strength grade

1.2.5 Type of timber

(5 × 1) (5)

[15]

QUESTION 2

2.1 Give ONE word or term for each of the following descriptions. Write only the word or term next to the question number (2.1.1–2.1.10) in the ANSWER BOOK.

2.1.1 The process of cutting metal using different saws

2.1.2 The process of cutting metal into different sizes and shapes using a torch flame

2.1.3 The process of neatening edges of a metal

2.1.4 The process of cutting metal sheets accurately

2.1.5 The process of perforating (making row of holes) a metal sheet

2.1.6 The process of strengthening metal by straining it beyond its yield point

2.1.7 The process of changing metal properties by heat treatment

2.1.8 The process of stretching metal by means of tensile forces

2.1.9 The process of cutting metal with a plasma torch

2.1.10 The technology that uses a laser to cut through a metal

(10 × 1) (10)

2.2 Name and explain THREE different methods of cutting metals. (6)

2.3 Distinguish between a *pure metal* and *alloy*, and give examples of each as commonly used in construction. (4)

[20]

QUESTION 3

3.1 Differentiate between *mild steel* and *high-tensile* steel. (4)

3.2 Explain the following terms that apply when steel is under stress.

3.2.1 Plastic deformation

3.2.2 Yield point

3.2.3 Elastic deformation

(3 × 2) (6)

3.3 Show by means of a sketch the following steel sections:

3.3.1 I-section

3.3.2 Lipped angle

3.3.3 H-section

(3 × 2) (6)

3.4 Discuss the role of the handing-over certificate after completion of the steel structure. (4)

[20]

QUESTION 4

4.1 Briefly discuss the following methods of preventing steel corrosion:

4.1.1 Drainage design

4.1.2 Painting

4.1.3 Hot-dip galvanising

4.1.4 Electroplating

4.1.5 Inherent corrosion resistance

(5 × 2) (10)

4.2 List THREE groups of stainless-steel alloys. (3)

4.3 State TWO methods you can use to protect steel from corrosion. (2)

[15]

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QUESTION 5

5.1 Write the following abbreviations in full:

5.1.1 AWS

5.1.2 WPS

5.1.3 SMAW

5.1.4 FCAW

5.1.5 SAW

5.1.6 MMAW

(6 × 1) (6)

5.2 List FIVE advantages of welding. (5)

5.3 Explain the importance of welding in a well-ventilated workshop. (2)

5.4 Explain the main difference between brazing and soldering. (2)

5.5 Name FIVE different methods of fusion welding to join steel. (5)

[20]

QUESTION 6

How would you identify the following types of glass when installed in a building?

6.1 Toughened glass

6.2 Laminated glass

6.3 Wired glass

6.4 Float glass

6.5 Glazed glass

(5 × 2) **[10]**

TOTAL: 100

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